

Order Blattodea

Cockroaches

Cockroaches are leathery insects, typically with oval and flattened bodies that enable them to squeeze through tight spaces in search of food or to escape predators. They are sensitive to vibration and can quickly flee from danger. The head is often covered by a shieldlike pronotum and there are generally 2 pairs of wings. The order Blattodea comprises 9 families and about 4,600 species, of which less than 1 percent are pests; the rest are useful scavengers in many habitats. The pest species thrive in warm conditions and where there is poor hygiene and sanitation, and can carry many disease-causing organisms on their bodies.

Adults may produce sexual pheromones to attract mates. The females lay up to 50 eggs in 2 rows surrounded by tough, protective egg cases called oothecae. Metamorphosis is incomplete.

Family Blaberidae

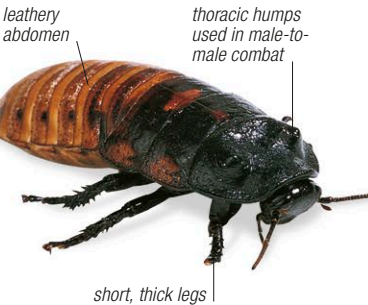
Live-bearing cockroaches

Occurrence 1,189 spp. in tropical regions; in caves, underground



These large insects, usually measuring 1–2¼ in (2.5–6 cm) in length, often have well-developed, pale brown wings with dark markings. In many species, however, the females are wingless and burrow under wood and stones. The pronotum may be very wide. During courtship, some species use sounds to communicate with a mate. All species are ovoviparous and give birth to live young. The egg sacs

are fully extruded from the end of the abdomen, rotated, and then drawn back inside to be brooded within the body of the female. Adults and nymphs are mainly saprophagous or eat dung.



GROMPHADORHINA PORTENTOSA
Better known as the *Madagascan hissing cockroach*, this insect is 2¼–3¼ in (6–8 cm) long, and startles predators by squeezing air out of its spiracles to produce a loud hiss.

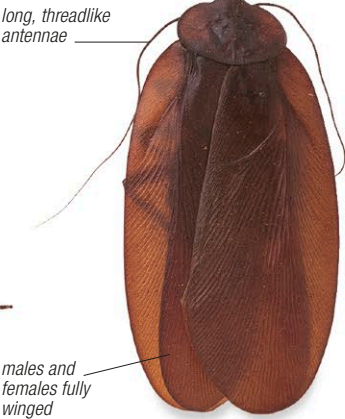
Family Blattellidae

Blattellids

Occurrence 2,240 spp. worldwide in warmer areas; in woodland litter, debris, garbage dumps, buildings



Many of these fully winged species are pale brown but a few are olive-green. Their body length varies from ⅝ to as long as 4 in (0.8–10 cm). Adults and nymphs are scavengers. Females carry their egg cases, which protrude from their bodies, until the eggs are about to hatch. Thousands of eggs may be produced in a lifetime.



MEGALOBLATTA LONGIPENNIS
The long-winged great cockroach from Peru, Ecuador, and Panama is the largest winged cockroach in the world. It has a wingspan of up to 8 in (20 cm).

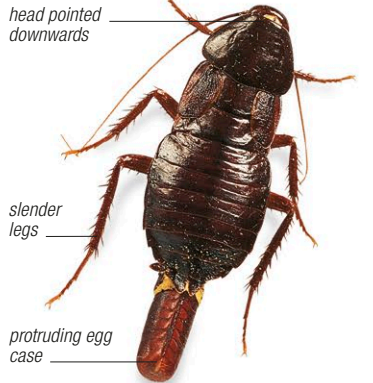
Family Blattidae

Common cockroaches

Occurrence 652 spp. mainly in tropical and subtropical regions; common in warehouses, sewers, garbage dumps



Most species of common cockroaches are brown or reddish or blackish brown with markings. They are ¾–1¼ in (2–4.5 cm) long, very active, and can run very fast or fly. Some species emit repellent chemicals that can cause skin rashes. Females can produce up to 50 egg cases, each containing 12–14 eggs, which are stuck to concealed surfaces. Both adults and nymphs are nocturnal and saprophagous.



BLATTA ORIENTALIS
The oriental cockroach is a common domestic pest. Like many species, it has spread from its native region by traveling on board ships.

Order Isoptera

Termites

Like ants (see p.576)—with which they are sometimes confused—termites are social insects, living in colonies that can be over a million strong. There are about 3,000 species, in 7 families, and most live in the tropics, where they build impressive nests. They are usually pale and soft, with chewing mouthparts and short antennae, but their body form varies with their role in the colony. Workers are small, while soldiers have large heads and jaws. Queens, which lay all the colony's eggs, can be as big as a human finger. Some termites feed on wood or plants, but others eat fungi that they raise in underground “gardens.” Although important in food chains as nutrient recyclers, many are destructive pests. Metamorphosis is incomplete.

Family Rhinotermitidae

Subterranean termites

Occurrence 360 spp. worldwide in warmer regions; in soil or in damp wood touching ground



As adults and nymphs, subterranean termites feed mainly on wood, digesting it with the help of microbes that they harbor in their intestines. They are voracious feeders, and in warm parts

of the world are highly damaging pests. Generally cream or light brown, they are typically ¼–¾ in (6–8 mm) long.



RETICULITERMES LUCIFUGUS
This termite has distinctive antennae. It makes its nest under the ground, inside damp wood, and is commonly found throughout southern Europe.

Family Termitidae

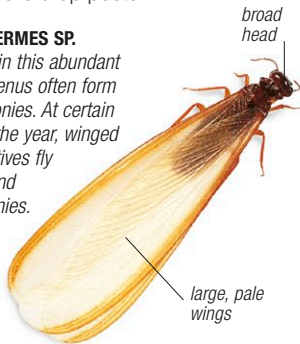
Higher termites

Occurrence 2,021 spp. worldwide; in trees, soil, underground



Three-quarters of the world's termite species belong to this family. Highly variable, both in appearance and habits, they range in length from ⅝–1½ in (4–14 mm). Queens are much bigger than the workers and soldiers, and may produce thousands of eggs a day. Nests vary from small structures built in trees, to vast earthen towers, with underground “fungus gardens.” Many species are crop pests.

MACROTREMES SP.
Termites in this abundant African genus often form giant colonies. At certain times of the year, winged reproductives fly off to found new colonies.



GLOBITERMES SULPHUREUS
This Southeast Asian pest makes thick-walled nests partly underground. The nests' soldiers can “explode” to cover intruders in a sticky fluid.



TRINERVITERMES TRINERVOIDES
The snouted harvester termite lives in pastures in central and southern Africa. It is eaten by the aardwolf (see p.209)—an animal that has evolved immunity to its defensive secretions.